Engine Model(s): PT6A

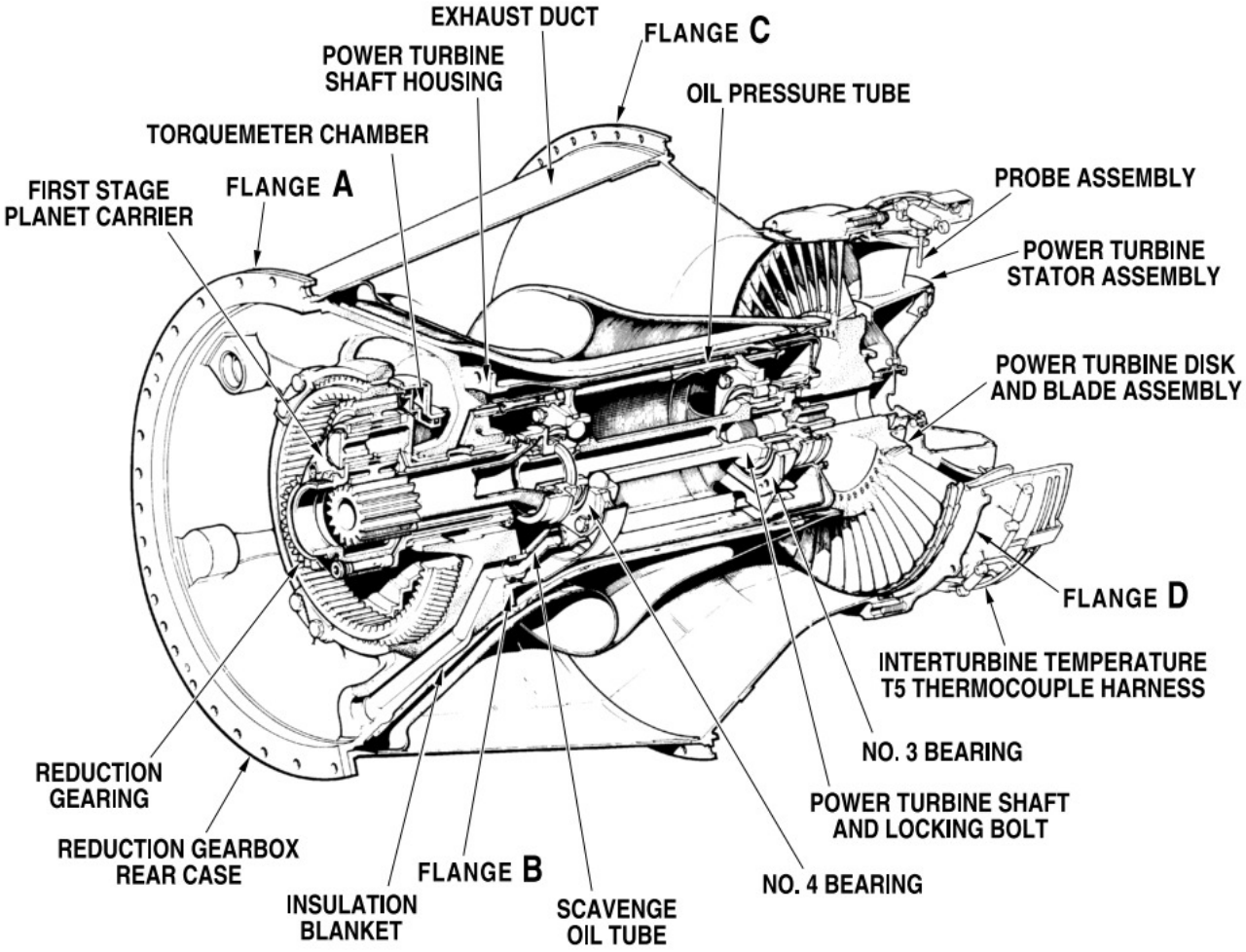
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POWER TURBINE - DESCRIPTION AND OPERATION

1. Description and Operation (Ref. Fig. 1)

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Figure 1 Power Section (Typical)



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The power turbine consists of a disk and blade assembly which is mounted on the rear end of the power turbine shaft and retained by a retaining bolt and keywasher. The disk incorporates 41 blades which differ from those of the compressor turbine blades in that they are cast with notched and shrouded tips. Each blade is located in a fir-tree serration in the circumference of the turbine disk and retained by a rivet. The disk and blade assembly rotates within a double knife-edge shroud to form a continuous seal at the blade tips when the engine is running.

The power turbine disk incorporates a flange for disk balancing. The required number of classified weights and rivets is determined during assembly balancing procedures.

The turbine disk and blade assembly is splined to the power turbine shaft and secured by a single retaining bolt and keywasher. A master spline makes sure that the turbine assembly may be removed and reinstalled in the same position to retain its original balance.

The turbine rotor balancing assembly is supported and secured in the power turbine shaft housing by the No.3 and No.4 bearings. The No.3 bearing is a roller-type while the No.4 bearing is a ball-type; both bearings have flanged outer races which permit bolting within the centerbore of the shaft housing. The No.3 bearing inner race and ball cage are positioned on the power turbine shaft by the power turbine disk and air seal runner at the rear and a shoulder on the shaft at the front. The No.4 bearing has a split inner race that is stacked between a shoulder power turbine shaft and a positioning ring and coupling which, in turn, are secured by a keywasher and spanner nut. The front half of the inner race has a puller groove for removal.

The power turbine shaft housing consists of a fabricated steel cylindrical housing attached at its front end to the rear case of the reduction gearbox. A labyrinth-type air seal, secured at the rear of the housing by a retaining ring, prevents lubricating oil from leaking into the power turbine area. An oil transfer tube which has four nozzles provides lubrication requirements for the No. 3 and No. 4 bearings. In normal operation, scavenge oil drains forward into the reduction gearbox sump; however, a scavenge tube is provided at the bottom of the shaft housing to scavenge the housing when the engine is in a nose-up attitude.

The power turbine shaft housing, together with the shaft assembly, is protected from the hot exhaust gases, when the engine is running, by an insulation blanket interposed between the inner wall of the exhaust duct and the power turbine shaft housing and associated rear case of the reduction gearbox.

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